

2020
Higher School Certificate
Year 11 Examination

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- NESA-approved calculators may be used
- Draw diagrams using pencil
- A Formulae Sheet is provided
- Write your student number and/or name at the top of every page

Total marks – 100

Section I – Pages 2–10

20 marks

Attempt Questions 1–20

Allow about 30 minutes for this section

Section II – Pages 11–29

80 marks

Attempt Questions 21–45

Allow about 1 hour and 30 minutes for this section

This paper MUST NOT be removed from the examination room

STUDENT NUMBER/NAME:

STUDENT NUMBER/NAME:

Section I

20 marks

Attempt Questions 1–20

Allow about 30 minutes for this section

Select the alternative A, B, C or D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
1				
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- 1 Which of the following measurements has 3 significant figures?

A. 15.08 cm
B. 500 m
C. 2.045 L
D. 4.02×10^4 km

- 2 A container has only two different coloured discs, green and white, in the ratio 5:3.

What is the probability that a white disc will be randomly selected from the container?

A. $\frac{3}{5}$
B. $\frac{3}{8}$
C. $\frac{1}{3}$
D. $\frac{2}{3}$

- 3 A school zone has a speed limit of 40 km/h.

What is this speed in m/s?

A. 1.1
B. 6.7
C. 11.1
D. 66.7

- 4 The data set has 5 scores: {10, 12, 12, 15, 18}

When another score of 12 is added to the set, which of these measures would change?

A. Mode
B. Range
C. Median
D. Mean

- 5 Marcella works casually and earns \$38.50 for her normal hourly rate.

For a particular week, she works 30 hours at the normal rate and 4 hours overtime at the time-and-a-half-rate.

What did Marcella earn in wages during the week?

- A. \$924
- B. \$1309
- C. \$1366.75
- D. \$1386

- 6 If J and k are related by the formula: $J = \sqrt{4k^3}$, what is the value of J if $k = 4$?

- A. 4
- B. 8
- C. 16
- D. 64

- 7 One quarter of a cup of almonds weighs approximately 35 g.

100 g of almonds contains 575 calories.

How many calories would one full cup of almonds contain?

- A. 615
- B. 805
- C. 915
- D. 975

- 8 On which lines are errors introduced into Amy's solution to the equation:
 $5x - 3(2x + 7) = 15$?

$$5x - 3(2x + 7) = 15$$

$$5x - 6x + 21 = 15 \quad \text{Line 1}$$

$$-x + 21 = 15 \quad \text{Line 2}$$

$$-x = 6 \quad \text{Line 3}$$

$$x = -6 \quad \text{Line 4}$$

- A. Line 1 only
 B. Line 1 and Line 2 only
 C. Line 1 and Line 3 only
 D. Line 1, Line 2 and Line 3
- 9 The diagram shows the graphs of two lines: L_1 with equation $V = 3t + 20$ and L_2 .

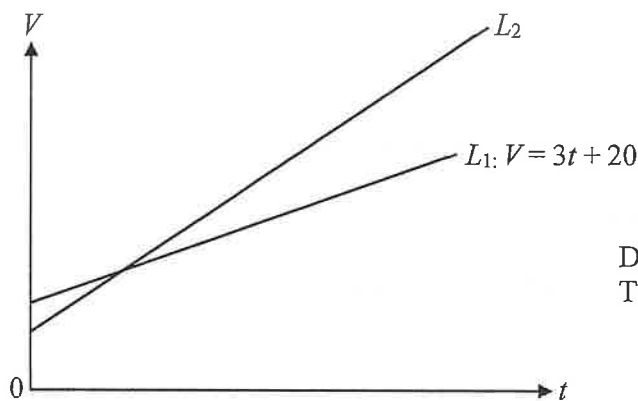


DIAGRAM NOT
TO SCALE

- Which of the following equations could represent line L_2 ?
- A. $V = 4t + 15$
 B. $V = 4t + 30$
 C. $V = 2t + 15$
 D. $V = 2t + 30$
- 10 What is the percentage error (+/-) in the measurement of 6.25 litres?
- A. 0.005
 B. 0.01
 C. 0.05
 D. 0.08

STUDENT NUMBER/NAME:

- 11 Two cities have coordinates $(10^{\circ}\text{N}, 40^{\circ}\text{E})$ and $(22^{\circ}\text{S}, 40^{\circ}\text{E})$.

What are the coordinates of a city halfway between these two cities?

- A. $(32^{\circ}\text{N}, 80^{\circ}\text{E})$
B. $(12^{\circ}\text{S}, 40^{\circ}\text{E})$
C. $(6^{\circ}\text{S}, 40^{\circ}\text{E})$
D. $(16^{\circ}\text{S}, 80^{\circ}\text{E})$
- 12 The heights of a number of plants grown from seeds are recorded in the stem-and-leaf plot. (23 | 4 refers to a height of 23.4 cm)

23	4
24	1 6 9
25	0 3 3 5 6
26	2 4 5

What is the lower quartile of the recorded heights of the plants?

- A. 24.6
B. 24.65
C. 24.75
D. 25.3
- 13 What is the perimeter of the figure shown below?

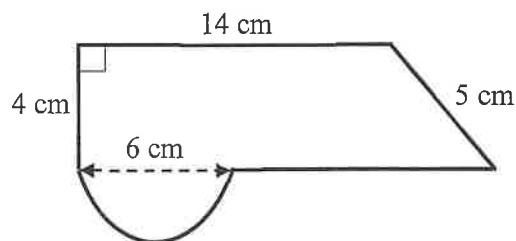
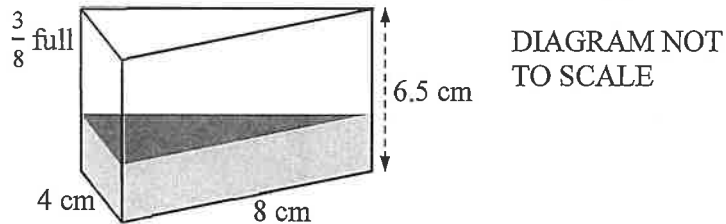


DIAGRAM NOT
TO SCALE

- A. $3\pi + 23$
B. $3\pi + 34$
C. $6\pi + 23$
D. $6\pi + 31$

- 14 A right triangular prism (open at the top) with base dimensions of 4 cm and 8 cm and height 6.5 cm is $\frac{3}{8}$ full of water.



How many millilitres of water are in the prism? (Note: $1 \text{ cm}^3 = 1 \text{ mL}$)

- A. 6
B. 12
C. 39
D. 78
- 15 James took out comprehensive insurance for his new car which costs \$2100 for the first 12 months.

If James does not make any claims within the first 12 months, he is entitled to a discount of 15% off his yearly premium for the following 12 months (assuming no increase in premium).

James has not made a claim during the first 12 months and elects to pay his premium monthly for the next 12 months.

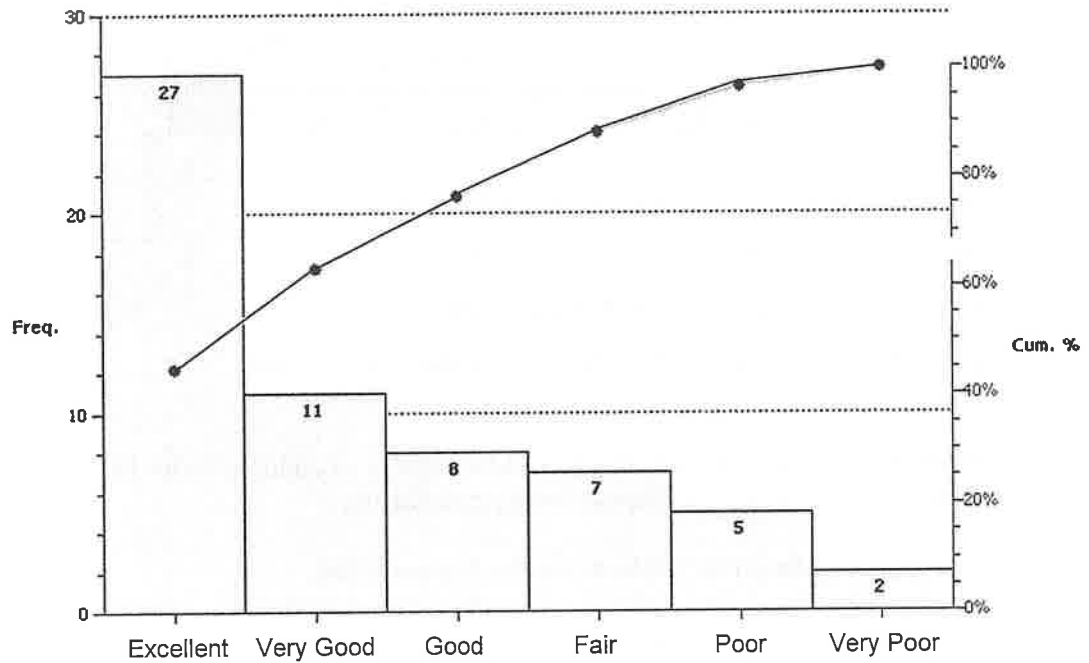
What will be his monthly premium?

- A. \$112.92
B. \$123.75
C. \$148.75
D. \$156.25

STUDENT NUMBER/NAME:

- 16 The Pareto chart shows the number of customers who responded to a survey relating to a company's services.

The survey asked customers to rate the services from "Excellent" to "Very poor".

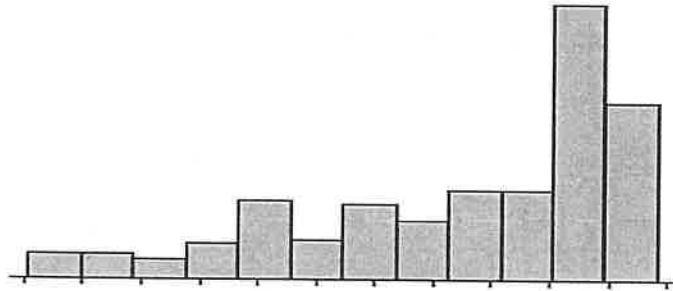


Approximately what percentage of customer responses rated the company's services in the bottom three categories?

- A. 10
- B. 23
- C. 30
- D. 40

STUDENT NUMBER/NAME:

- 17 Which statement gives a correct description of the data distribution shown?



- A. Negatively skewed with the mode greater than the mean
B. Positively skewed with the mean greater than the mode
C. Positively skewed with the mode greater than the mean
D. Negatively skewed with the mean greater than the mode
- 18 Students in a class were asked to record the number of children under 18 years old living at home (including themselves) in their family.

The following frequency table shows the data collected.

<i>Number of children</i>	<i>Frequency</i>
1	6
2	8
3	
4	2
5	0

From the data it was determined that the relative frequency of those families with 3 children was $\frac{1}{5}$.

What number is missing from the frequency table?

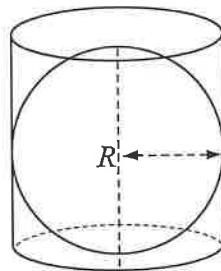
- A. 1
B. 2
C. 3
D. 4

- 19 Damian purchased a guitar for $\$M$ and later sold it making a 25% profit.

What did Damian sell his guitar for?

- A. $\frac{5M}{4}$
- B. $\frac{M}{4}$
- C. $\frac{4M}{5}$
- D. $\frac{M}{5}$

- 20 A sphere is placed inside a cylinder so that it touches the top and bottom as well as the sides of the cylinder.



The cylinder and sphere both have radius R .

What is the ratio of the volume of the cylinder to the volume of the sphere?

- A. 3 : 2
- B. 5 : 3
- C. 5 : 4
- D. 4 : 3

Section II**80 marks****Attempt Questions 21–45****Allow about 1 hour and 30 minutes for this section**

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the end of the examination. If you use this space, clearly indicate which question you are answering.

Question 21 (1 mark)**Marks**

A person's income can be described as low, medium or high.

1

Write the data classification for this description.

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Question 22 (1 mark)What is the standard deviation of this data set of weights?
(Give your answer correct to TWO decimal places.)**1**

62 g, 69 g, 74 g, 76 g, 89 g, 96 g, 102 g

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Question 23 (2 marks)

Sydney has a Coordinated Universal Time (UTC) of +11 and San Francisco has a UTC of -8.

2When it is 1:15 pm on Sunday 5th of the month in San Francisco, what time and date is it in Sydney?

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STUDENT NUMBER/NAME:

Question 24 (2 marks)

Marks

One grain of rice has a mass of 0.00171 grams.

2

Approximately how many grains of rice are required to make 1.5 kg? (Write your answer in scientific notation with TWO significant figures.)

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Question 25 (2 marks)

A person's height (h), mass (m) and body mass index (b) can be related by the formula.

2

$$b = \frac{m}{h^2}$$

Claire is 1.7 m tall and had her body mass index recorded on two separate occasions at 21.5 and 24.5.

What was Claire's average weight (kg) from the two recordings?

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Question 26 (2 marks)

Government stamp duty on new motor vehicle purchases is based on the following rates:

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- 3% of the purchase amount up to \$45 000
- + 5% of the purchase price over \$45 000

What would be the amount of stamp duty payable on a new motor vehicle with a purchase price of \$62 500?

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Question 27 (2 marks)**Marks**

The smallest of 5 numbers in a data set is 5.

2

If the mode, median and range all have a value of 8, write a possible data set satisfying these conditions.

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Question 28 (3 marks)

Madison weighs 42 kg and has consumed 3 standard drinks in the last 2 hours.

(a) Use the formula: $BAC = \frac{10N - 7.5H}{5.5M}$

2

Where: N is the number of standard drinks
 H is the number of hours drinking
 M is the person's weight

to calculate Madison's Blood Alcohol Content (BAC) correct to TWO decimal places.

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(b) Madison stops drinking at 10:00 pm.

1

The number of hours (D) she has to wait before her BAC will likely reach zero is given by the formula:

$$D = \frac{BAC}{0.015}$$

At what time will Madison's BAC likely reach zero?

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STUDENT NUMBER/NAME:

Question 29 (3 marks)

Marks

The formula: $C = \frac{5(F - 32)}{9}$

3

is used to convert Fahrenheit temperatures (F) (in degrees) into Celsius temperatures (C) (in degrees).

Write this formula with F as the subject.

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Question 30 (2 marks)

Solve the equation: $\sqrt{7 - 3p} = 2\sqrt{p}$

2

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STUDENT NUMBER/NAME:

Question 31 (3 marks)

Marks

The following table shows the fortnightly Youth Allowance to people who are studying, training or in an apprenticeship.

3

<i>Status</i>	<i>Fortnightly allowance</i>
Under 18, living at home	\$249.20
Under 18, living away from home	\$455.20
18 and over, living at home	\$299.80
18 and over, living away from home	\$455.20

The annual youth allowance is reduced by \$0.20 for every \$1 a youth's parents' annual income is over \$53 728.

Tony is in an apprenticeship, is 18 years old and living away from home.

If Tony's parents earn a combined annual income of \$78 100, determine the fortnightly youth allowance Tony is entitled to.

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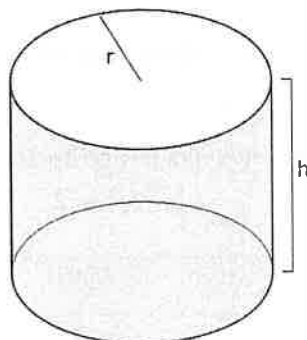
STUDENT NUMBER/NAME:

Question 32 (2 marks)

Marks

The radius of a closed cylindrical can is 10 cm and its height is 20 cm.

2



What is the surface area of the can correct to the nearest whole number?

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Question 33 (2 marks)

The semicircular design shown is part of a pattern on a wallpaper feature.

2

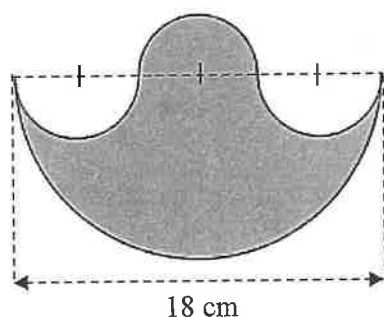


DIAGRAM NOT TO SCALE

Calculate the area of the design giving your answer correct to the nearest whole number.

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Question 34 (3 marks)**Marks**

The amount of water in a swimming pool which is full to capacity of 65 000 litres, reduces by 2.5% per day through evaporation.

3

After 2 days of evaporation, the owner added water to the pool which brought it back to full capacity.

What percentage increase in water (correct to one decimal place) did the owner add after the second day?

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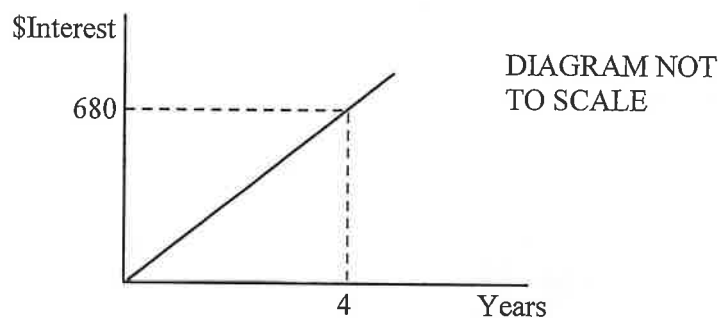
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Question 35 (2 marks)

The graph shows the interest earned over a 4-year period on a lump sum investment.

2

If the annual flat rate of interest paid was 5%, what amount was invested?

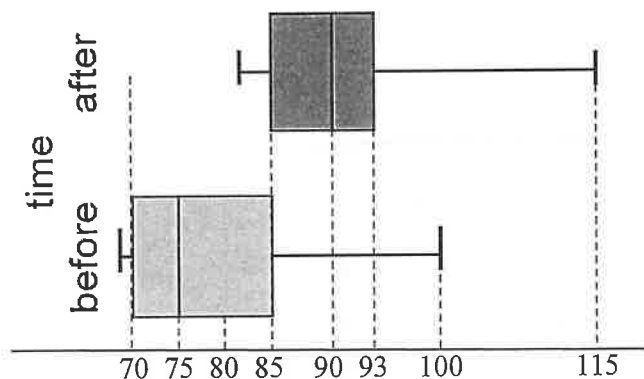
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Question 36 (5 marks)**Marks**

The parallel box and whisker plots below show the changes in the pulse rates (beats/min) of 40 people before and after exercise at a local fitness centre.



- (a) What change occurred in the median pulse rate after exercise? 1

- (b) What change occurred in the interquartile range after exercise? 1

- (c) After exercise, how many people had a pulse rate beyond the upper quartile? 1

- (d) Explain with calculations why, after exercise, pulse rates recorded as 115 beats per minute, could be considered outliers for the recorded data. 2

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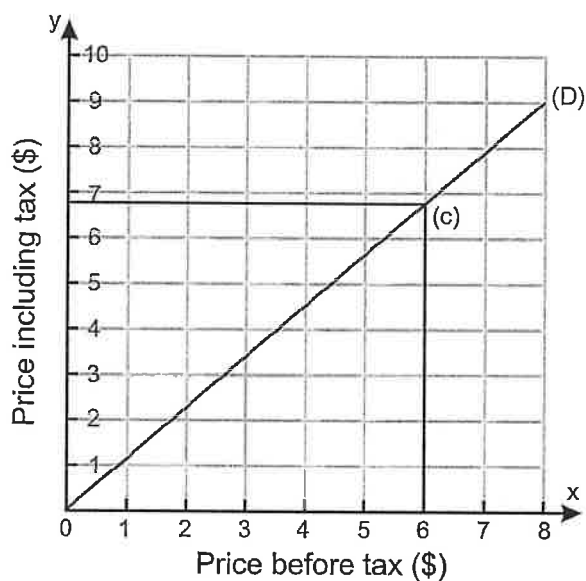
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Question 37 (6 marks)**Marks**

The linear graph shown, compares the prices of items before and after a tax was added.



- (a) The point D on the graph has coordinates (8, 9).

2

By using this point, explain why the equation of the graph is $y = 1.125x$.

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- (b) Explain why the rate of tax added to the prices of the items is 12.5%..

1

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- (c) Write the coordinates of the point (c) shown on the graph.

1

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- (d) The price after the tax was added to an item is \$150.

2

What amount of tax was added to the original price?

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Question 38 (6 marks)**Marks**

Brendon's annual gross income for the past financial year was \$112 525.00 and he paid a total of \$30 250 in tax (PAYG).

The following table lists Brendon's allowable tax deductions and the amounts he can deduct from his gross income.

<i>Allowable deductions</i>	<i>Amount claimed</i>
Private Health Fund contributions	\$135 per fortnight
Union Fees	\$220 annually
Home office equipment \$2850	12.5% annual depreciation
Professional magazine subscription	\$15.00 monthly
Vehicle expenses Travel 2820 km	\$0.37 per km

- (a) Show that Brendon's taxable income for the past financial year was \$107 215 to the nearest dollar. 2

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- (b) Brendon has to pay a Medicare Levy based on 2% of his taxable income. 1

Using (a) what is the Medicare Levy amount Brendon needs to pay.

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Question 38 continues on the next page

STUDENT NUMBER/NAME:

Question 38 (continued)

Marks

- (c) The following tax table gives the rates of tax Brendon will use to determine if he is entitled to a tax refund or whether he will have a tax debt for the past financial year.

3

<i>Taxable income \$</i>	<i>Tax payable</i>
0–18 200	nil
18 201–37 000	19 cents for each \$1 over \$18 200
37 001–87 000	\$3572 plus 32.5 cents for each \$1 over \$37 000
87 001–180 000	\$19 822 plus 37 cents for each \$1 over \$87 000
180 001 and over	\$54 232 plus 45 cents for each \$1 over \$180 000

Use the tax table and your answers to (a) and (b) to determine whether Brendon is eligible for a tax refund or whether he will have a tax debt for the past financial year and write down the amount.

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End of Question 38

STUDENT NUMBER/NAME:

Question 39 (3 marks)

Marks

Use the formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

3

to determine the two values of x_2 , if $x_1 = -3$, $y_2 = -2$, $y_1 = 4$ and $d = 10$.

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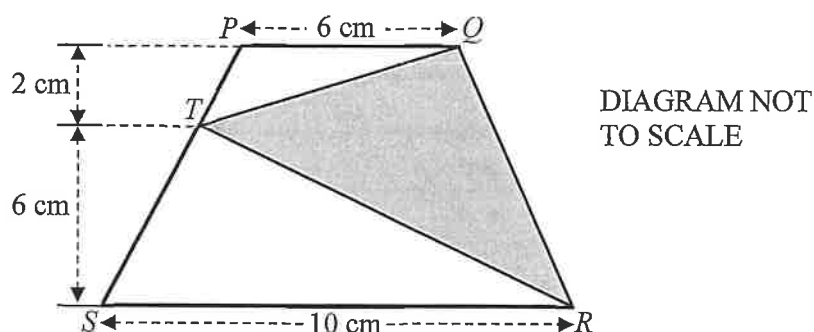
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Question 40 (5 marks)**Marks**

A triangular piece of paper QRT is placed on top of a white piece of cardboard in the shape of a trapezium $PQRS$ as shown in the diagram.

The edge QR of both the triangle and trapezium are aligned.



- (a) Show the area of the trapezium shaped cardboard $PQRS$ is 64 cm^2 . 1

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- (b) Use the result of (a) to show that the area of the triangular piece of paper QRT is 28 cm^2 . 2

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- (c) What percentage of the cardboard does the paper cover? 2

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Question 41 (4 marks)**Marks**

The length of the arc PQ of this shape is 12π cm.

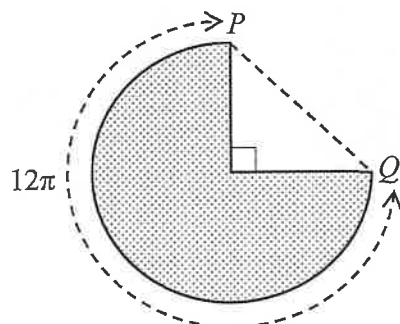


DIAGRAM NOT
TO SCALE

- (a) Show that the radius of the circle is 8 cm.

2

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- (b) What is the length of the chord PQ of the circle correct to TWO decimal places?

2

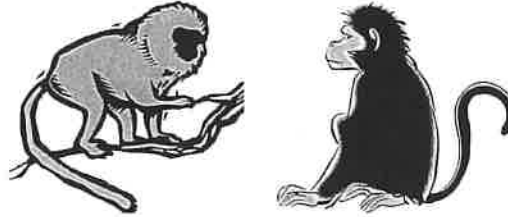
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Question 42 (5 marks)**Marks**

There are two different endangered species of monkey living in a heritage listed forest.



40% of the monkeys are of species A and 60% are of species B .

Two monkeys are randomly selected for vaccination.

What is the probability that:

- (a) Both monkeys are from species A ? 1

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- (b) The monkeys are of different species? 2

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- (c) At least one of the monkeys is of species B ? 2

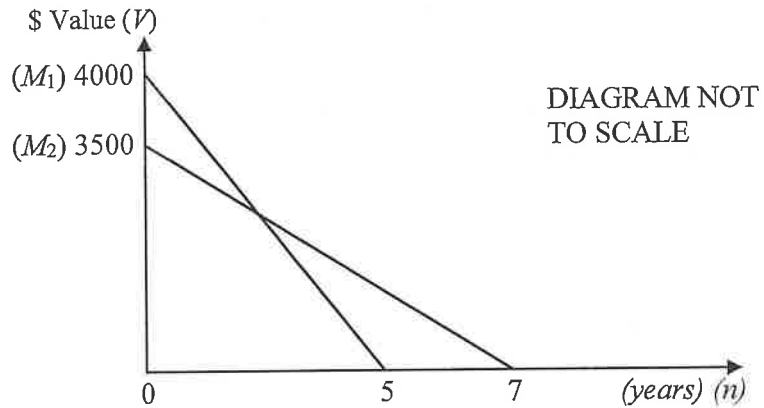
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Question 43 (5 marks)**Marks**

The graph shows the values of two office machines (M_1) and (M_2) when purchased new.



The straight line method of depreciation is applied to determine the value of the machines after each year of use.

- (a) Which machine has the highest annual rate of depreciation? 1
-
- (b) What is the remaining value of machine M_2 after machine M_1 has no value? 1
-
- (c) After how many years and months will the machines have the same value? 3

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STUDENT NUMBER/NAME:

Question 44 (3 marks)

Marks

The petrol tank in a motorist's car has a capacity of 60 litres.

3

It was calculated that 90 km of range was left before the tank is empty. The motorist fills the tank at a cost of \$1.72 per litre.

If the car's average fuel consumption is 8.5 L/100 km, how much (to the nearest dollar) would it cost the motorist to fill the tank.

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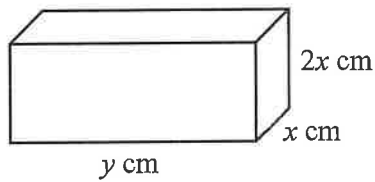
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Question 45 (6 marks)**Marks**

A rectangular prism has length y cm, width x cm and height $2x$ cm, as shown.



The sum of all the edges of the prism is 36 cm.

- (a) Show that $y = 9 - 3x$.

2

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- (b) Show that the volume of the prism can be given by: $18x^2 - 6x^3$ cm³.

2

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- (c) Explain why the length of the prism can only have values between 0 and 9 cm.

2

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End of paper

STUDENT NUMBER/NAME:

Section II – Extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a full page of primary-ruled paper. It features approximately 20 horizontal dashed lines spaced evenly down the page, providing a guide for handwriting practice. The paper is otherwise blank, with no margins, text, or other markings.

**Mathematics Standard 1**
Mathematics Standard 2**REFERENCE SHEET****Measurement****Limits of accuracy**

Absolute error = $\frac{1}{2} \times \text{precision}$

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

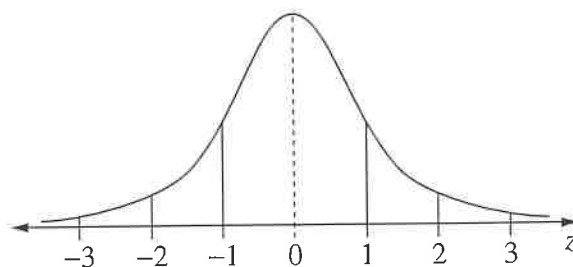
An outlier is a score

less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

$$z = \frac{x - \mu}{\sigma}$$

Normal distribution

- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

